

Impact of Utility-Scale Solar Development on Land Use and Carbon Dynamics in Massachusetts

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Summary of Works | July 2026

Link to the work samples

[Land Use Change Mapping](#) (Google Earth Engine API)

[Land Use Change Analysis](#) (Google Earth Engine Code/ GitHub)

[Statewide Carbon Modeling](#) (R Studio Code / GitHub)

Overview and Key Findings

Global efforts to achieve net-zero greenhouse gas emissions have led to rising demand for renewable energy sources aimed at reducing dependency on fossil fuels. Among renewable energy technologies, photovoltaic (PV) energy has experienced sustained global expansion and is widely recognized as a key strategy for reducing emissions from fossil fuel-based electricity generation (Creutzig et al., 2017). In the United States, utility-scale PV development has grown rapidly over the past two decades, with installations exceeding 1 MW increasingly occurring across agricultural and forested landscapes (Seel et al., 2024). Utility-scale solar facilities require substantial land area, and their deployment may cause land use and land cover change (LULCC), particularly in forested regions (Ong et al., 2013). Solar developments have been associated with changes in albedo, land surface temperature, soil properties, hydrology, and land-atmosphere interactions (Hernandez et al., 2014). Of particular concern is deforestation for solar infrastructure, where emissions associated with clearing aboveground biomass (AGB) may offset part of the intended carbon benefits of renewable electricity. As renewable energy projects expand, their climate benefits can no longer be assumed and must be evaluated through explicit and integrated carbon modeling. If terrestrial carbon emissions associated with solar development are not evaluated alongside carbon savings from avoided fossil fuel emissions, renewable deployment strategies may overestimate or mischaracterize their net carbon benefits. In regions where forests function as significant carbon sinks, understanding the balance between immediate carbon losses and long-term emission reductions becomes essential for aligning renewable energy deployment with broader net-zero climate goals. The analysis **finds that utility-scale solar development in Massachusetts has generated substantial net carbon benefits, but forest conversion measurably delays carbon payback and reduces cumulative mitigation, particularly as the regional electricity grid decarbonizes.** More Specifically:

- **Approximately half of utility-scale solar development between 2005 and 2024 occurred on previously forested land.** The estimated utility-scale solar footprint was $41.5 \pm 9.9 \text{ km}^2$, of which approximately 51% was installed following forest clearing.
- **Land disturbance extended beyond the areas occupied by solar arrays.** When adjacent clearing associated with site development was included, total solar-associated deforestation reached $34.6 \pm 7.4 \text{ km}^2$.

- **Deforestation created both immediate and persistent carbon costs.** Cumulative deforestation-related carbon fluxes over 2005–2054 totaled 373.4 ± 90.9 Gg C, including immediate biomass removal, delayed emissions from biomass decay, and the loss of potential forest carbon sequestration.
- **Utility-scale solar ultimately produced a substantial statewide net carbon benefit, but this benefit was not realized immediately.** The cumulative statewide carbon balance did not reach breakeven until approximately 2021 and remained net negative (relative to the atmosphere) from 2022 onward. By 2054 (the end of the 30 year lifetime of solar installed by 2024), cumulative net carbon emissions are estimated at -2.91 ± 0.77 Tg C.
- **Life cycle and land use emissions offset a meaningful share of the carbon savings from solar generation.** Over 2005–2054, photovoltaic life-cycle emissions were equivalent to approximately 22% of cumulative avoided grid emissions, while deforestation-related emissions accounted for approximately 9%.
- **Deforestation became a rapidly growing component of the carbon cost of solar development as the electricity grid decarbonization.** For representative projects, deforestation-related emissions increased from 11.9% of cumulative carbon savings for a system installed in 2005 to 20.3% for one installed in 2024. In contrast, the share attributable to PV life-cycle emissions increased only from 22.9% to 24.9%, reflecting continued improvements in PV manufacturing and performance. As avoided emissions decline with grid decarbonization, deforestation is expected to consume an increasingly large share of the net mitigation benefit.

These key findings and the methods used to produce them are elaborated below (1.2 Summary of Methods and 1.3 Summary of Results), with additional context and explanation of the rationale, nuances, and assumptions of the findings.

1. Summary of Methods

This study used a framework to quantify where utility-scale solar development occurred, estimate the associated terrestrial and energy-system carbon fluxes, and integrate these components into annual and cumulative net carbon emissions. The analysis covered utility-scale solar development in Massachusetts between 2005 and 2024 and projected carbon pathways through 2054 with 30 years of operations.

1.1 Mapping Solar Development and LULCC

Utility-scale solar installations and associated land use changes were mapped using all available Landsat time series observations from 2005 through 2024. The Continuous Change Detection and Classification (CCDC) was used to identify the timing and magnitude of changes in vegetation, exposed soil, built surfaces, albedo, and land surface temperature. These temporal-change features were then grouped into spatially coherent objects using Simple Non-Iterative Clustering and classified with machine learning. The classification distinguished solar installations developed after forest clearing, installations on non-forested land, and forest clearing adjacent to solar facilities. The resulting error matrix was used to calculate statistically unbiased area estimates and 95% confidence intervals for solar development and associated deforestation.

1.2 Estimating Energy-System Emissions and Carbon Savings

Greenhouse gas emissions from utility-scale PV systems were calculated by a cradle-to-grave Life Cycle Assessment (LCA) that included module and balance-of-system manufacturing, construction, operation, maintenance, and end-of-life processes. However, deployment year is an important determinant of PV life cycle carbon intensity and must be explicitly represented when evaluating emissions from solar systems installed across multiple decades. To account for this, published life cycle estimates were modeled as a function of cumulative global installed PV capacity using a learning-curve approach. This provided time-variant carbon intensity values as an input of the carbon model. Annual solar electricity generation was simulated with the PVWatts model in the National Renewable Energy Laboratory's System Advisor Model (SAM). Site-level generation estimated were aggregated using 5,000 weighted bootstrap samples to represent both the spatial distribution of solar development and regional variability in solar electricity production in Massachusetts. Avoided emissions were evaluated under the observed and projected annual average carbon intensity of the ISO New England (ISO-NE) electricity grid as it accounts for continued regional grid decarbonization.

1.3 Integrated Carbon Bookkeeping and Uncertainty Propagation

Figure 1 represents the integrated carbon accounting model used in this study, including how terrestrial carbon dynamics and energy-system emissions are combined to estimate net carbon emissions of utility-scale solar development.

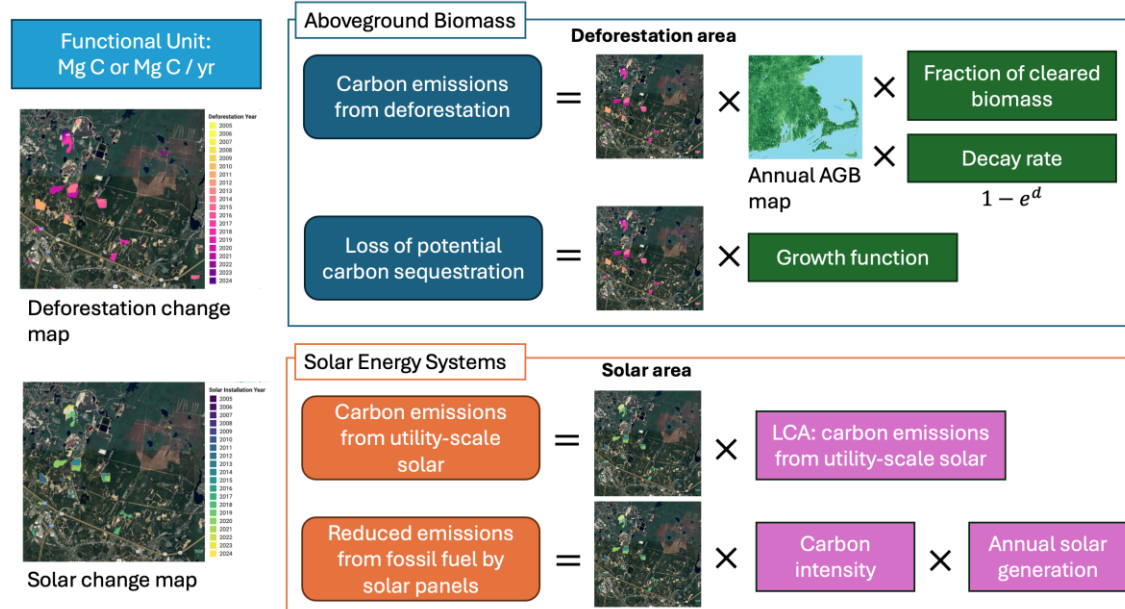


Figure 1: Schematic overview of the integrated carbon bookkeeping model combining land use change, biomass dynamics, and solar energy systems to estimate annual and cumulative net carbon emissions from utility-scale solar development.

Terrestrial carbon component was estimated by combining the area estimate from satellite time series analysis with AGB density derived from the U.S Forest Service TreeMap product. Forest growth rates were estimated from Forest Inventory and Analysis (FIA) data. Carbon removed during deforestation was represented through three carbon pathways: immediate emissions from biomass removal, delayed emissions from softwood and hardwood decay pools, and the continuing loss of potential carbon

sequestration deforestation. All terrestrial and energy-system components were converted to common units of carbon mass per year and combined as:

$$\text{Net carbon emissions} = \text{deforestation-related emissions} + \text{PV life cycle emissions} - \text{avoided grid emissions}$$

Uncertainty was propagated through 5,000 Monte Carlo simulations. Each iteration sampled from distributions representing uncertainty in solar and deforestation area, AGB, forest growth, biomass decay parameters, and PV electricity generation. The resulting distributions were used to calculate annual and cumulative mean carbon fluxes and corresponding 95% confidence intervals.

2. Summary of Results

2.1 Land Use Change Associated with Utility-Scale Solar Development

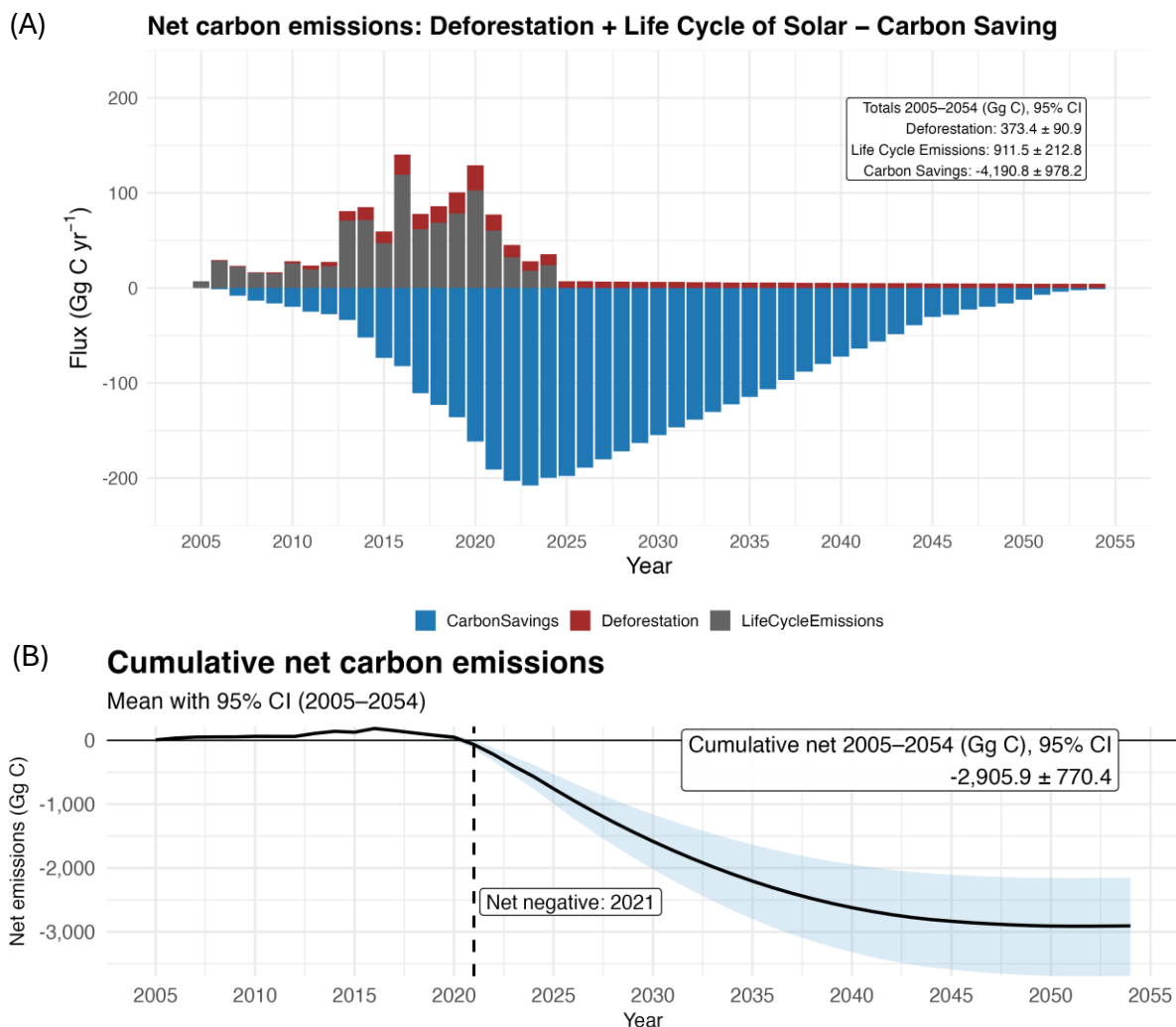


Figure 2: Net carbon emissions of utility-scale solar development in Massachusetts, 2005–2054. (A) Annual carbon fluxes from deforestation, photovoltaic life-cycle emissions, and avoided ISO New England grid emissions. (B) Cumulative net carbon emissions with the 95% confidence interval. Statewide carbon breakeven occurred in 2021, and cumulative net emissions reached $-2,905.9 \pm 770.4$ Gg C by 2054.

This study developed a time-varying carbon accounting framework that integrates land use change, terrestrial carbon dynamics, PV life cycle emissions, electricity generation, and grid decarbonization. From the satellite time series analysis, the estimated utility-scale solar footprint in Massachusetts reached $41.5 \pm 9.9 \text{ km}^2$ between 2005 and 2024. Approximately 51% of this area ($21.1 \pm 5.9 \text{ km}^2$) was installed following direct forest clearing. An additional $13.5 \pm 5.0 \text{ km}^2$ of adjacent forest was cleared, increasing total solar-associated deforestation to $34.6 \pm 7.4 \text{ km}^2$. Together, solar arrays and adjacent clearing affected 55 km^2 , demonstrating that current inventories or mapping efforts based only on panel footprints may understate the full land use impact of solar development.

Figure 2 summarizes the resulting statewide carbon balance. Over 2005–2054, avoided ISO-NE grid emissions totaled $-4.19 \pm 0.98 \text{ Tg C}$. These savings were partially offset by $0.91 \pm 0.21 \text{ Tg C}$ of PV life-cycle emissions and $0.37 \pm 0.09 \text{ Tg C}$ of deforestation-related emissions, equivalent to approximately 22% and 9% of cumulative avoided emissions, respectively. After accounting for both upfront carbon emissions, cumulative net carbon flux reached $-2.91 \pm 0.77 \text{ Tg C}$ by 2054. The benefit was substantial but delayed: the statewide cumulative balance did not reach carbon breakeven until approximately 2021 and remained net benefit thereafter.

At the project level, sites involving forest clearing required approximately five to six years to reach carbon payback under the ISO-NE grid displacement scenario. The time-varying analysis also shows that grid decarbonization increases the relative importance of land use emissions. For a representative system installed with deforestation in 2005, deforestation-related emissions were equivalent to 11.9% of cumulative carbon savings, compared with 20.3% for a system installed with deforestation in 2024—an increase of 8.4 percentage points. Over the same period, the share attributable to PV life cycle emissions increased only from 22.9% to 24.9%, even reflecting continued improvements in PV manufacturing and performance. As the electricity grid becomes cleaner, each unit of solar generation avoids fewer emissions, causing forest conversion to consume an increasingly large share of the remaining mitigation benefit.

These findings have direct implications for renewable energy planning. Prioritizing previously disturbed, developed, or otherwise lower-carbon lands can reduce terrestrial carbon losses, shorten carbon payback, and preserve a larger share of carbon benefit from renewable solar. These siting considerations will become increasingly consequential as the carbon intensity of displaced electricity declines, because each unit of solar generation will avoid fewer emissions while emissions from deforestation remain comparatively persistent. Despite the observed land use impacts, the approximately 55 km^2 of associated statewide solar developments was projected to generate net carbon savings equivalent to the 30-year carbon uptake of approximately $1,064 \text{ km}^2$ of Massachusetts forest. This comparison illustrates the substantial climate benefit of solar electricity relative to its geographic footprint, but it does not imply that deforestation is necessary or justified.

Finally, net carbon balance represents only one dimension of environmental sustainability. This assessment did not quantify soil-carbon losses, biodiversity impacts, habitat fragmentation, hydrological changes, microclimatic effects, or other ecosystem services affected by forest conversion. Overall, utility-scale solar development in Massachusetts generated substantial long-term carbon reductions, but the magnitude and timing of those benefits depended on where projects were built, when they were installed, and which electricity sources they displaced.